# Lecture 2: Introduction to Cryptography and Symmetric Key Encryption

**ITDS271: Computer and Communication Security**

*Dolvara Gunatilaka, PhD.*

# Topics

- **Introduction to Cryptography and Crtyptanalysis**
- Cryptographic Algorithms
- Classical Ciphers
  - Substitution Ciphers
  - Transposition Ciphers
- Modern Ciphers
  - DES & 3-DES

# Definition and Terms: Cryptology

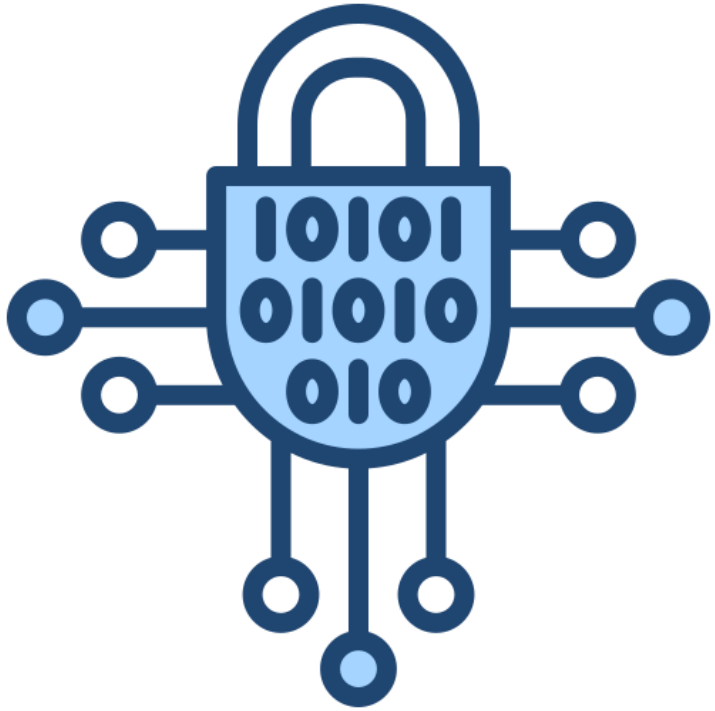
Mahidol University

**Cryptology**

- **Cryptography**
  An area of study concerning various techniques used for **encryption**
- **Cryptanalysis**
  The art of **breaking/cracking** (deciphering) the code without being told the key

# Cryptography

- The art of "**secret writing**"
- A way to transform one data into an unintelligible from.
  - Transform information (*plaintext*) into an unreadable format (*ciphertext*) and back again,
  - Ensure that only authorized parties can understand it.
- Enable confidentiality, integrity, authenticity, and non-repudiation
- Cryptography can be reversible or non-reversible



# How Crptography Provides Security

- **Confidentiality**
  - Allow only authorized users to access information
- **Integrity**
  - Trust the information has not been altered
- **Authentication**
  - Verify who the sender was and trust the sender is who they claim to be
- **Non-repudiation**
  - Ensure that sender or receiver cannot deny that a message was sent or received
- **Access control**
  - Restrict availability to information

# Cryptography = "Secret Writing"

- **Early Ciphers (Simple Encryption):** aims to conceal written communication
  - Methods that obscure the *meaning* of a message
  - E.g., Caesar cipher (shifting letters), Atbash cipher (reversing the alphabet), or the Spartan Scytale (transposition cipher using a rod)
- **Steganography:** the practice of concealing a message within another message or physical object → ***secret hiding***
  - Hidden message indiscernible to the naked eye or an unsuspecting observer
  - Need a carrier media, e.g., photo, audio, video, etc.
- **Modern Encryption:** use complex *mathematical* algorithms and computational power to perform encryption

# Example 1: Ancient Cipher

## Atbash Cipher

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | B | C | D | E | F | G | H | I | J | K | L | M |
| Z | Y | X | W | V | U | T | S | R | Q | P | O | N |
| N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
| M | L | K | J | I | H | G | F | E | D | C | B | A |

● ALPHABET ● ATBASH

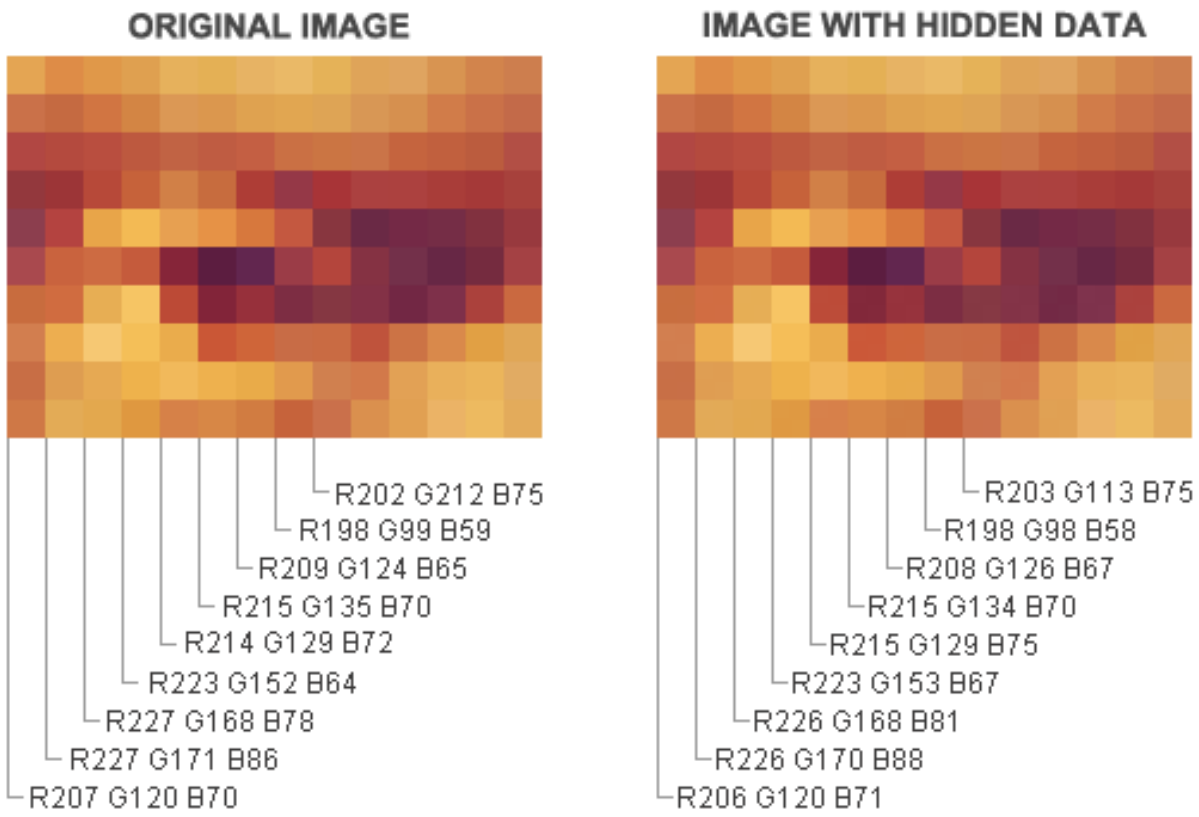
Plaintext = COMPUTER

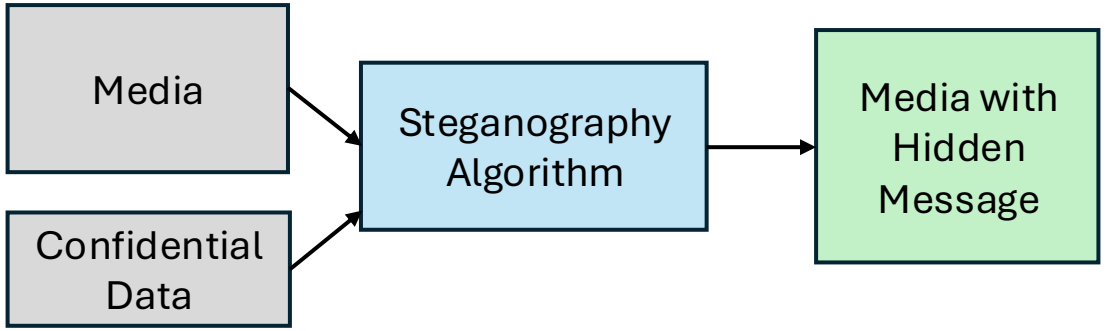
Ciphertext = _____

## Scytale Cipher

Image: https://en.wikipedia.org/wiki/Scytale

# Example 2: Steganography

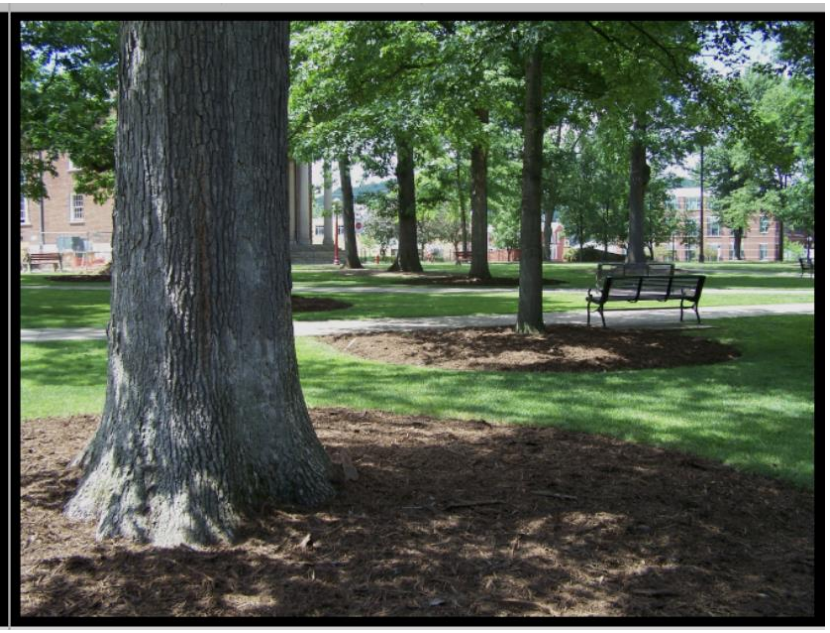



- An image file contains a binary representation of the color or light intensity of each pixel.
- To do steganography, simply change value of least significant bit(s) of the image to the information we want to hide.

# Example 3: Steganography

Picture of some trees

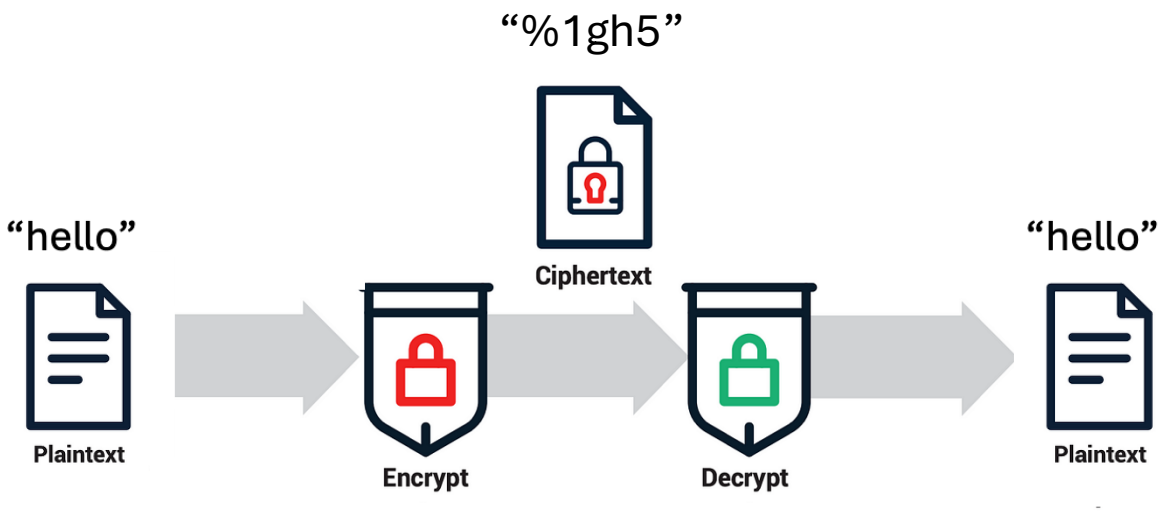


Picture of some tress + tar.gz file of the source code of this repository: https://github.com/gregfrazier/steganography-junk

Hard to tell/notice by human perceptions whether there is secret information embedded inside or not

# Cryptography: Encryption and Decryption

- **Encryption:** the process of converting plaintext into ciphertext
- **Decryption:** The process of converting plaintext into ciphertext
- Rely on ***cryptographic algorithm*** and a ***key***
- **Plaintext:** The original, readable information or message.
- **Ciphertext:** The encrypted, unreadable form of the information.



# Cryptanalysis

*Study and practice of analyzing or breaking cryptographic systems — in other words, it's the process of trying to* ***defeat encryption*** *without having access to the secret key*

- **Counterpart to cryptography!**
- Goal: to recover the key <u>not just the message</u>
- General approaches:
  - ***Cryptanalytic attacks***: Use *mathematical analysis* and knowledge of how the cipher works to find weaknesses.
  - ***Brute-force attack***: Try all possible keys

# Cryptanalysis Types

Types of attack are classified by how much the attacker knows about the message being transmitted.

| Type of Attack | Known to Cryptanalyst |
|---|---|
| Ciphertext Only | ■ Encryption algorithm<br>■ Ciphertext |
| Known Plaintext | ■ Encryption algorithm<br>■ Ciphertext<br>■ One or more plaintext–ciphertext pairs formed with the secret key |
| Chosen Plaintext | ■ Encryption algorithm<br>■ Ciphertext<br>■ Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key |
| Chosen Ciphertext | ■ Encryption algorithm<br>■ Ciphertext<br>■ Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key |
| Chosen Text | ■ Encryption algorithm<br>■ Ciphertext<br>■ Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key<br>■ Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key |

# Cryptanalysis: Brute-Force

- Try <u>every possible key</u> until the correct one is found
- Trial-and-error method that relies on *computational power* and the number of possible keys
- On average *half of all possible* keys must be tried before success

| Key Size (bits) | Number of Alternative Keys | Time required (at 1 decryption/μs) | Time required (at $10^6$ decryptions/μs) |
|---|---|---|---|
| 32 | $2^{32} = 4.3 \times 10^9$ | $2^{31}$ μs = 35.8 minutes | 2.15 milliseconds |
| 56 | $2^{56} = 7.2 \times 10^{16}$ | $2^{55}$ μs = 1142 years | 10.01 hours |
| 128 | $2^{128} = 3.4 \times 10^{38}$ | $2^{127}$ μs = $5.4 \times 10^{24}$ years | $5.4 \times 10^{18}$ years |
| 168 | $2^{168} = 3.7 \times 10^{50}$ | $2^{167}$ μs = $5.9 \times 10^{36}$ years | $5.9 \times 10^{30}$ years |
| 26 characters (permutation) | $26! = 4 \times 10^{26}$ | $2 \times 10^{26}$ μs= $6.4 \times 10^{12}$ years | $6.4 \times 10^6$ years |

# Topics

- Introduction to Cryptography and Crtyptanalysis
- **Cryptographic Algorithms**
- Classical Ciphers
  - Substitution Ciphers
  - Transposition Ciphers
- Modern Ciphers
  - DES & 3-DES

# Classification of Cryptographic Systems

- **CRYPTOGRAPHY**
  - **ENCRYPTION**
    - **Symmetric-key**
      - Block & Stream Ciphers
      - Substitution & Transposition
    - **Asymmetric-key**
      - RSA
      - Elliptic Curve Cryptography (ECC)
  - **1-WAY HASH**
    - MD5, SHA1-1
    - SHA-256, SHA-3

- Type of operations
- Number of keys
- The way the plaintext is processed

# Encryption

*Mathematical functions that work with a key*

- Security of data relies on:
  - Strength of the algorithm (i.e., cipher)
  - Secrecy of the key
- Usually, the key is kept as a secret
- But the encryption algorithm is known to public
- The same plaintext data is encrypted into different ciphertext with different keys

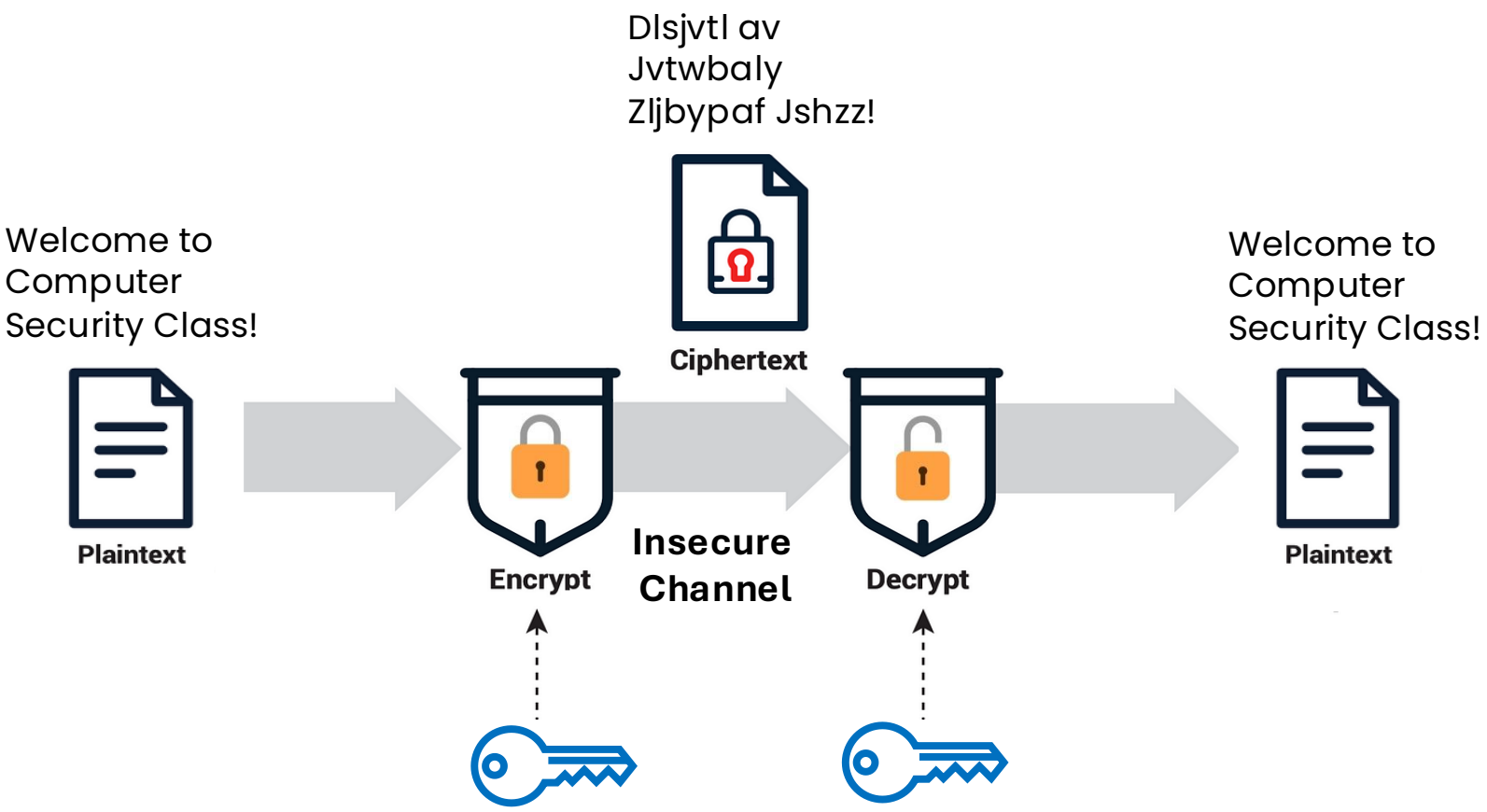
- Encryption algorithm are called "**Cipher**"
- Strength of a cipher depends on
  - Key size → Key space
    - 128-bit key vs 256-bit key
  - Algorithm Complexity
    - Still follow Kerkhoff's principle → ***open design***

***Encryption is used for securing communication and storage, authentication, integrity check.***

# Symmetric Key Encryption (1)

- Encryption and decryption use the **same set of parameters** to control the transformation
- Conventional / **private-key** / single-key
- Sender and recipient share a **common key**
- All classical encryption algorithms are **symmetric-key encryption**
- Computationally fast, require less computing resources (compared to **asymmetric-key** approach)
- Very efficient if implement correctly, think about AES!

# Symmetric Key Encryption (2)




**Shared secret-key**

(distributed to both parties before communication)

# Symmetric Key Encryption (3)

There are **two requirements** for secure use of symmetric-key encryption:

1. **Strong encryption algorithm**
   - Assume encryption algorithm is known (open)
2. **Secret key known only to sender /receiver**
   - Imply a secure channel is used to distribute the key

**Classical methods**

- Substitution: one letter is exchanged for another
- Transposition: the order of letters is rearranged
- Product system: combination of both methods

**Modern methods** – use complicated combination of classical methods and some mathematical computation

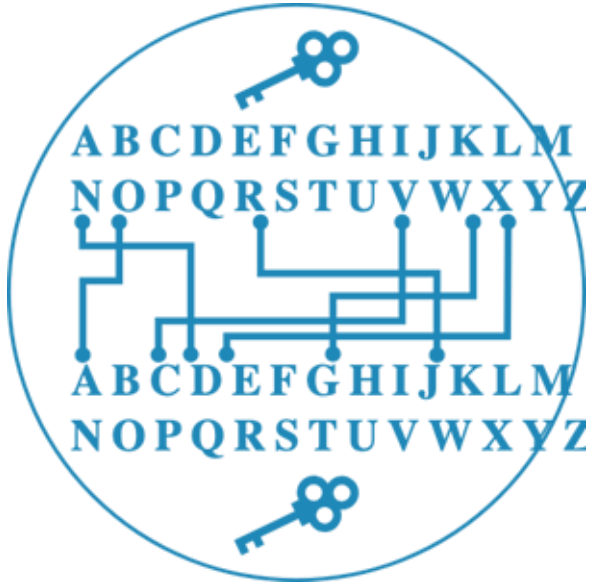
- Data Encryption Standard (DES) – obsoleted
- Advanced Encryption Standard (AES) - currently used

# Topics

- Introduction to Cryptography and Crtyptanalysis
- Cryptographic Algorithms
- **Classical Ciphers**
  - **Substitution Ciphers**
  - **Transposition Ciphers**
- Modern Ciphers
  - DES & 3-DES

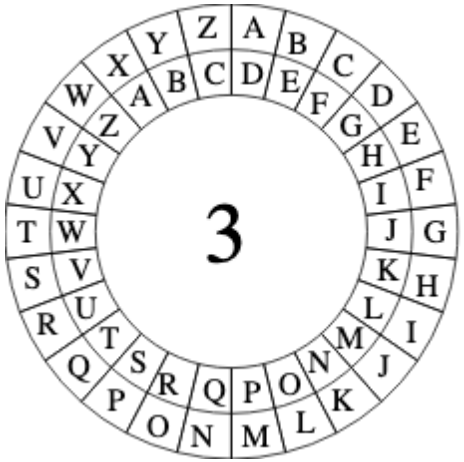
# Substitution Cipher

- Letters of plaintext are **replaced/substitued** by other characters, i.e., letters/ numbers/ symbols
- Including substituting bit patterns within the plaintext
- Examples:
  - **Monoalphabetic** Cipher: Caesar Cipher, Simple Substitution
  - **Polyalphabetic** Cipher: Vigenère and Vernam ciphers, One-Time Pad



# Monoalphabetic: Caesar Cipher (1)

- Earliest/oldest substitution cipher used by Julius Caesar
- Replace each letter with the letter standing ***k*** places further down the alphabet
- ***k*** is the secret key in this algorithm
- Have a loop property, so the <u>letter following Z is A</u>.
- According to the original use of Caesar Cipher, ***k* = 3**




**Example (key = 3):**

| | |
|---|---|
| I love Mahidol | (Plaintext) |
| L oryh Pdklgro | (Ciphertext) |

We can mathematically express Caesar Cipher as:

$$c = E(k, p) = (p + k) \bmod 26$$

$$d = D(k, c) = (c - k) \bmod 26$$

# Monoalphabetic: Caesar Cipher (2)

**Cryptanalysis of Caesar Cipher:**

- **26** possible ciphers
- Easy for brute force search

***Far from secure!***

`L oryh Pdklgro`

| | |
|---|---|
| +3 | I love Mahidol |
| +10 | B ehox Ftabwhe |
| +11 | A dgnw Eszavgd |
| +6 | F ilsb Jxefali |
| +24 | N qtaj Rfmnitq |
| +25 | M pszi Qelmhsp |
| +23 | O rubk Sgnojur |
| +9 | C fipy Gubcxif |
| +2 | J mpwf Nbijepm |
| +21 | Q twdm Uipqlwt |
| +13 | Y belu Cqxyteb |
| +7 | E hkra Iwdezkh |
| +22 | P svcl Thopkvs |

| | |
|---|---|
| +19 | S vyfo Wkrsnyv |
| +18 | T wzgp Xlstozw |
| +14 | X adkt Bpwxsda |
| +17 | U xahq Ymtupax |
| +15 | W zcjs Aovwrcz |
| +4 | H knud Lzghcnk |
| +16 | V ybir Znuvqby |
| +1 | K nqxg Ocjkfqn |
| +5 | G jmtc Kyfgbmj |
| +20 | R uxen Vjqrmxu |
| +12 | Z cfmv Dryzufc |
| +8 | D gjqz Hvcdyjg |

# Monoalphabetic: Simple Substitution (1)

- Instead of shifting the alphabet
- Each letter of the plaintext maps to a different ciphertext letter
- Thus, the key length is 26 letters long
  - A total of 26! ($4 \times 10^{26}$) possible keys
- Is it secure? No!
  - Suffer from <u>language characteristics</u>

**Example:**

| Plain | A | B | C | D | E | F | G | H | I | J | K | L | M |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Cipher | X | F | A | C | P | T | K | W | U | G | Z | B | I |

| Plain | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Cipher | Q | E | D | H | J | Y | O | L | N | V | M | R | S |

**Plaintext:** **mypassword**

**Ciphertext:** **irdxyyvejc**

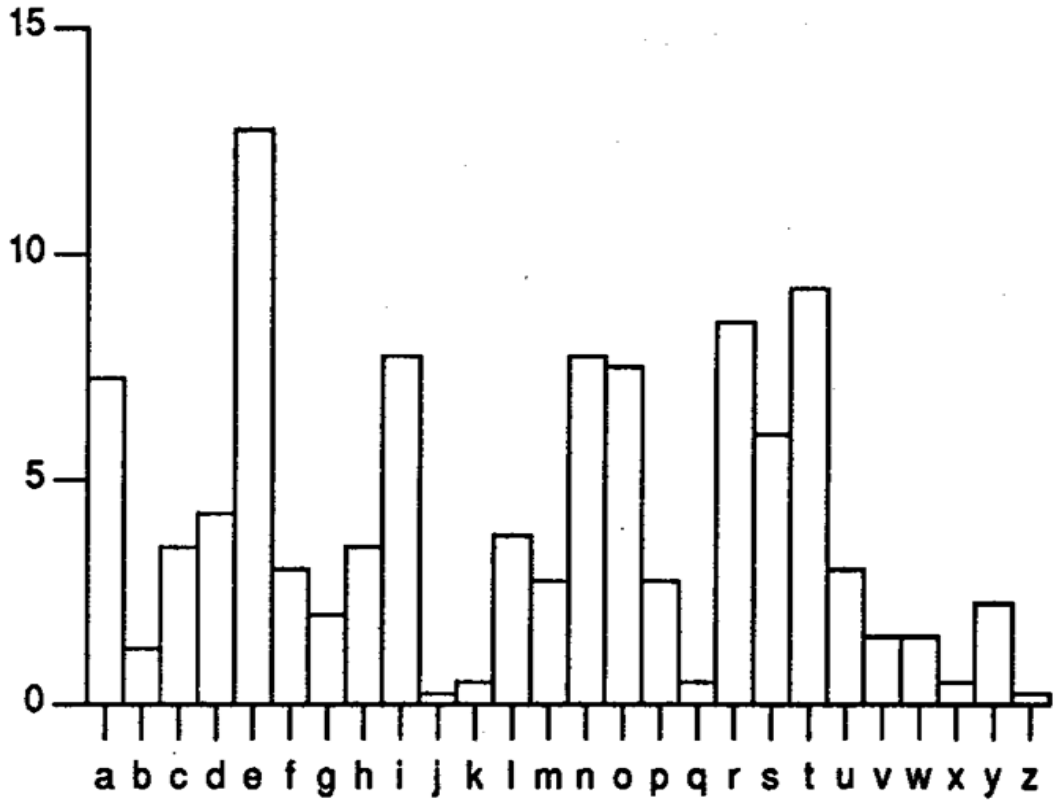
# Monoalphabetic: Large Redundancy

***Language Redundancy***

**Letters** are **<u>not equally used</u>**

- **'E'** is by far the most common letter used in English followed by **T, R, N, I, O, A, S**
- Digram (2 letters combination) → **TH**
- Trigram (3 letters combination) → **THE**

***Frequency analysis:*** is a way to break simple codes by looking at how often letters show up in the ciphertext




English character frequencies

# Frequency Analysis Example

```
UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZVUEPHZHMDZSHZOWSFPAPPDTSVPQ
UZWYMXUZUHSXEPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ
```

- Count <u>relative letter frequency</u>
- Guess **'P' → 'e'** and **'Z' → 't'**
- **'ZW' → 'th'** and **'ZWP' is 'the'**
- Proceed with trial and error, we will finally get:

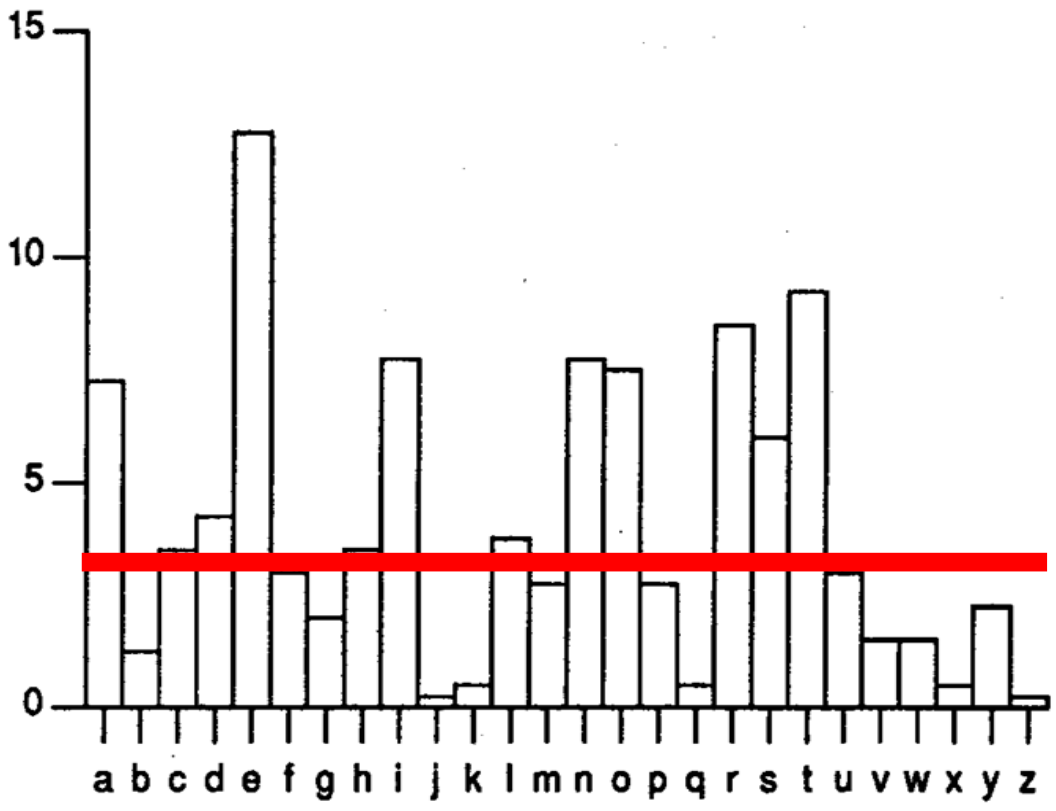
```
it was disclosed yesterday that several informal but direct
contacts have been made with political representatives of the
viet cong in moscow
```

Occurency and Frequency Analysis
1-grams

| ↑↓ | ↑↓ | ↑↓ | ↑↓ |
|---|---|---|---|
| P | 16× | 13.33% | |
| Z | 14× | 11.67% | |
| U | 10× | 8.33% | |
| S | 10× | 8.33% | |
| O | 9× | 7.5% | |
| M | 8× | 6.67% | |
| H | 7× | 5.83% | |
| E | 6× | 5% | |
| D | 6× | 5% | |
| V | 5× | 4.17% | |
| X | 5× | 4.17% | |
| W | 4× | 3.33% | |
| F | 4× | 3.33% | |
| Q | 3× | 2.5% | |
| T | 3× | 2.5% | |
| G | 2× | 1.67% | |
| B | 2× | 1.67% | |
| A | 2× | 1.67% | |
| Y | 2× | 1.67% | |
| I | 1× | 0.83% | |
| J | 1× | 0.83% | |

# Substitution: Polyalphabetic Cipher

- Improve security using multiple substitution cipher alphabets
- **Flatten** frequency distribution
- More complex than monoalphabetic ciphers.
- Resistant to simple frequency analysis because letter frequencies are obscured
- Examples**: Vigenère** and **Vernam ciphers**




English character frequencies

# Polyalphabetic Cipher: Vigenère

- Simplest polyalphabetic substitution cipher
- Effectively *multiple Caesar Ciphers*
- The key is a multiple-letter long → $K = k_1\ k_2 \ldots k_n$
- The keyword is repeated to match the length of the plaintext message

**Example:**

- **Plaintext: "we are discovered "**
- **Key: "deceptive"**
- we will have the ciphertext looks like this:

```
key:        deceptive|deceptive|deceptive
plaintext:  wearedisc|overedsav|eyourself
ciphertext: ZICVTWQNG|RZGVTWAVZ|HCQYGLMGJ
```

# Polyalphabetic Cipher: Vigenère

- Each letter is encrypted using <u>Caesar</u> cipher
  - Shift amount is determined by the corresponding letter of repeated keyword
- To find the shift value for a key letter,
  - Assign 'A' = 0, 'B' = 1, ..., 'Z' = 25
- Encryption formula for each character Ci
  - $\mathbf{C_i=(P_i+K_i)mod26}$

| a | b | c | d | e | f | g | h | i | j | k | l | m |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |

| n | o | p | q | r | s | t | u | v | w | x | y | z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |

```
key:        deceptive | deceptive | deceptive
plaintext:  wearedisc | overedsav | eyourself
ciphertext: ZICVTWQNG | RZGVTWAVZ | HCQYGLMGJ
```

# Polyalphabetic Cipher: Vigenère

- A plaintext letter can be transformed into multiple ciphertext letters
  - Not a 1-to-1 relationship, like in Caesar Cipher
- Letter frequency are more obscured
- However, some <u>periodic pattern</u> can still be observed

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| key | 3 | 4 | 2 | 4 | 15 | 19 | 8 | 21 | 4 | 3 | 4 | 2 | 4 | 15 |
| plaintext | 22 | 4 | 0 | 17 | 4 | 3 | 8 | 18 | 2 | 14 | 21 | 4 | 17 | 4 |
| ciphertext | 25 | 8 | 2 | 21 | 19 | 22 | 16 | 13 | 6 | 17 | 25 | 6 | 21 | 19 |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| key | 19 | 8 | 21 | 4 | 3 | 4 | 2 | 4 | 15 | 19 | 8 | 21 | 4 |
| plaintext | 3 | 18 | 0 | 21 | 4 | 24 | 14 | 20 | 17 | 18 | 4 | 11 | 5 |
| ciphertext | 22 | 0 | 21 | 25 | 7 | 2 | 16 | 24 | 6 | 11 | 12 | 6 | 9 |

```
key:        deceptive deceptive deceptive
plaintext:  wearedisc overedsav eyourself
ciphertext: ZICVTWQNG RZGVTWAVZ HCQYGLMGJ
```

# Polyalphabetic: Vernam

- Select the keyword as long as the plaintext
- <u>But, **key may repeated**</u>
- Invented by Gilbert Vernam, an engineer from AT&T, in 1918
- Convert an alphabet into a number, add it with a random number and mod by 26
- Also, support stream cipher where each bit of plaintext is XOR with the bit of the key.

| a | b | c | d | e | f | g | h | i | j | k | l | m |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |

| n | o | p | q | r | s | t | u | v | w | x | y | z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |

| Plaintext | V | E | R | N | A | M |
|---|---|---|---|---|---|---|
| Numeric Equivalent | 21 | 04 | 17 | 13 | 0 | 12 |
| + Random number (key) | 76 | 48 | 16 | 82 | 44 | 03 |
| Sum | 97 | 52 | 33 | 95 | 44 | 15 |
| mod 26 | 19 | 0 | 7 | 17 | 18 | 15 |
| Ciphertext | T | A | H | R | S | P |

# Perfect Substitution

- Use **random** distribution for many alphabets
- No predictable pattern for the choice of an alphabet
- A **perfect key**, an unlimited, non-repeating series of numbers
- Thus, the repeated plaintext would not be encrypted the same way twice, and the duplication of two ciphertext letters is almost accidental
- Some Efforts: Vernam Cipher → One-time pad (OTP)

**PERFECT**
**=!**
**PRACTICAL!**

# Substitution: One-Time Pad

- Secure to the degree that is considered **unbreakable!**
- The key is **truly random** and is only for **one-time use**
- The key is at least **as long as the plainttext** is used to encipher the message
- No pattern, trace left, since the key is purely random and has no periodic pattern → **perfect secrecy**

**Two fundamental difficulties in deploying One-Time Pad:**

- Safe key distribution
- Generating large quantities of random keys

# Transposition Cipher

- Now, considered **Transposition** or **Permutation** Ciphers
- Transform a plaintext message into a ciphertext by rearranging the letter order
- Without altering the actual letters within the plaintext
- Can be recognized using frequency analysis, since the output (ciphertext) still have the same frequency distribution as the original plaintext
- E.g., Rail Fence cipher, and row transposition

**Scytale**

# Transposition: Rail Fence

- Simplest form of transposition cipher
- Letters in plaintext are rewritten diagonally over the number of **rails/ key**
- Then, read the data row-by-row and use it as the ciphertext.
- Simple to implement
- Easy to break by trying different number of rails

**Plaintext: WE ARE DISCOVERED FLEE AT ONCE**

**Ciphertext: WECRLTEERDSOEEFEAOCAVDNNE**

**Rails = 3**

```
W . . . E . . . C . . . R . . . L . . . T . . . E
. E . R . D . S . O . E . E . F . E . A . O . C .
. . A . . . I . . . V . . . D . . . N . . . E . .
```

# Transposition: Row Transposition

- Write letters of the message out in rows over a specified number of columns
- Read the message off column by column
- The order of column read then become the **key** to the algorithm
- More complex than rail fence, but is still susceptible to frequency analysis

```
Key:        4 3 1 2 5 6 7
Plaintext:  A T T A C K P
            O S T P O N E
            D U N T I L T
            W O A M X Y Z

Ciphertext:
TTNAAPTMTSUOAODWCOIXKNLYPETZ
```

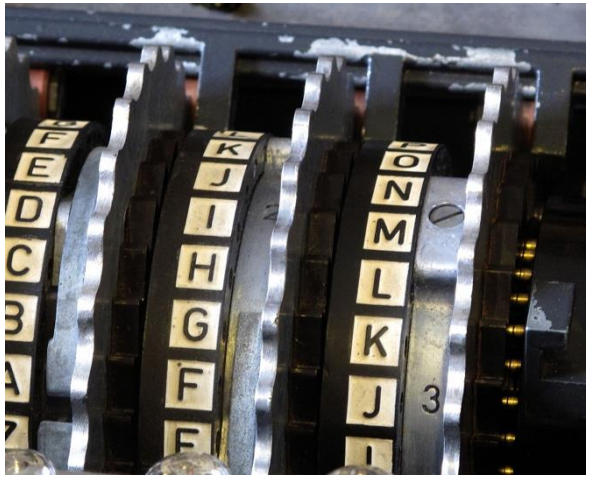
# Product Cipher

- Combining 2 substitution ciphers makes a …
  - more complex substitution cipher
- Combining 2 transposition ciphers makes a …
  - more complex transposition cipher
- However, combining a substitution cipher with transposition cipher makes a …
  - <u>much harder cipher</u>

*This is the bridge separating* ***<u>classical cipher</u>*** *from the* ***<u>modern one</u>***

# Rotor Machine

- **Electromechanical encryption devices** used mainly during the early- to mid-20th century to automate **substitution** ciphers
- Use a very complex substitution cipher
- Played a major role in military and diplomatic communications during WWII
- The most famous example is the German **Enigma machine**.



# Security of Encryption

## 1. Unconditionally Secure Algorithm

- No matter how much computing resources (power or time) is available
- One-Time Pad (OTP) is the only method that is considered **unconditionally secure**

## 2. Computationally Secure Algorithm

- Given the limited computing resources the cipher cannot be broken
- A cipher is conditionally is secured if
  - The cost of breaking cipher exceeds the value of information contained inside or
  - The time used to break the cipher is longer than the useful lifetime of the information

# Topics

- Introduction to Cryptography and Crtyptanalysis
- Cryptographic Algorithms
- Classical Ciphers
  - Substitution Ciphers
  - Transposition Ciphers
- **Modern Ciphers**
  - **DES & 3-DES**

# Modern Ciphers: Two Flavors of Encryption

## Block Cipher

- Encrypt data in fixed-size ***blocks*** (e.g., 64 bits, 128 bits).
- The same key is used for each block, but the encryption process is often made dependent on previous blocks
- **Analogy:** Encrypting a document page by page

DES | 3-DES | AES

## Stream Cipher

- Encrypt data ***bit by bit*** or byte by byte.
- Generate a pseudorandom keystream, which is then combined (typically XORed) with the plaintext
- **Analogy:** Encrypting a document letter by letter as it's being written.

RC4 | Salsa20 | ChaCha20

# Design of Modern Symmetric Key Encryption

1. Have a general **iterative block cipher** structure
   - with number of rounds
   - with substitutions/permutations controlled by key
2. Parameters
   - **Block size** and **key size**
   - Algorithm to generate subkey (used for each round)
   - Different round functions for each round
3. Fast software implementation and execution

# Modern Symmetirc Key Encryption Algorithms

| Algorithm | Year | Block Size (bits) | Key Size (bits) | Description |
|---|---|---|---|---|
| Lucifer | ~1970 | 64 | 128 | Developed by Horst Feistel at IBM; the first widely recognized modern block cipher. Uses Feistel network structure. |
| DES | 1977 | 64 | 56 | Adopted as a U.S. federal standard, based on IBM's Lucifer. A Feistel cipher with 16 rounds. |
| IDEA | 1991 | 64 | 128 | International Data Encryption Algorithm. |
| Blowfish | 1993 | 64 | 32 to 448 (variable) | A Feistel-like cipher with 16 rounds and variable key length. |
| RC5 | 1994 | Flexible (32, 64, or 128) | Flexible (0 to 2040) | Parameterized cipher with flexible block size, key size, and number of rounds. |
| Triple DES | ~1998 | 64 | 112 or 168 (effective) | Enhancement to DES designed to increase its key length and overcome brute-force vulnerability. |
| AES | 2001 | 128 | 128, 192, or 256 | Selected by NIST to replace DES/3DES. Based on the Rijndael algorithm. |

# Data Encryption Standard (DES)

- 1977 – 2001
- Was a most widely used block cipher in the world
- Adopted in 1977 by NBS (now NIST) – as ***FIPS PUB 46***

- IBM developed a **Lucifer cipher**
  - by the team led by Fiestel in late 60's
  - used 64- bit data blocks with 128-bit key
- Redeveloped as a commercial cipher with input from **NSA** and others
- 1973, NBS issued a request proposals for a national cipher standard
- IBM submitted their revised Lucifer cipher which was eventually accepted as **DES**

# Fiestal Cipher Structure

- Known as Feistel network
- **Round-based** Structure
  - Split plaintext into <u>two</u> halves and process them through multiple **rounds**
- Involve **substitution** and **permutation** using ***round function (F)*** and ***subkeys ($K_i$)***
  - Round function can be simple or complex
- Ciphers that use fiestal networ:
  - DES, Blowgish, Twofish, etc.

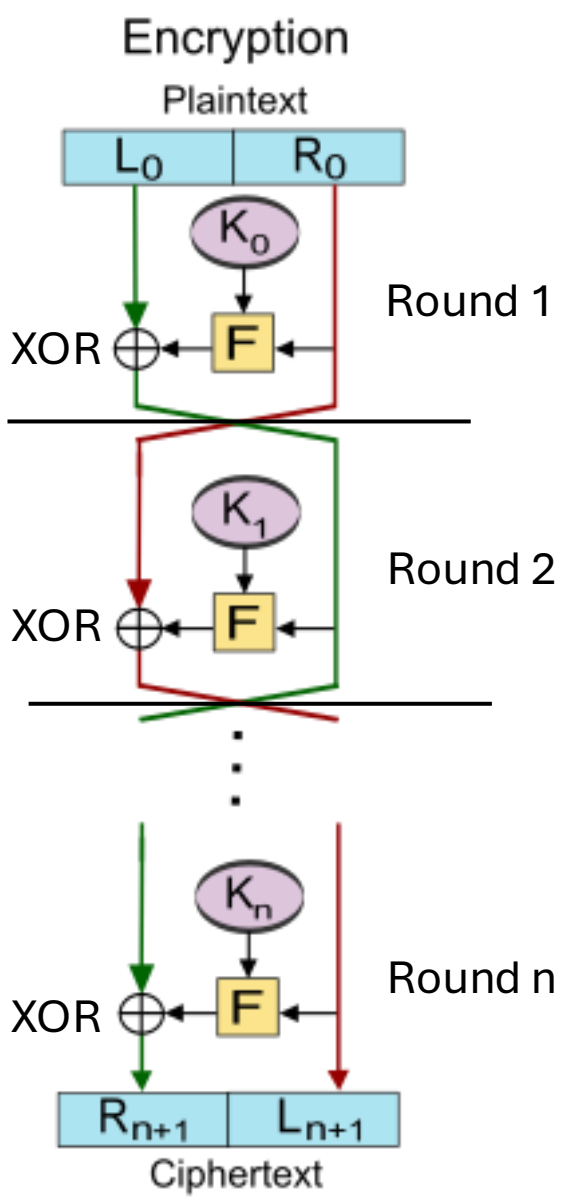
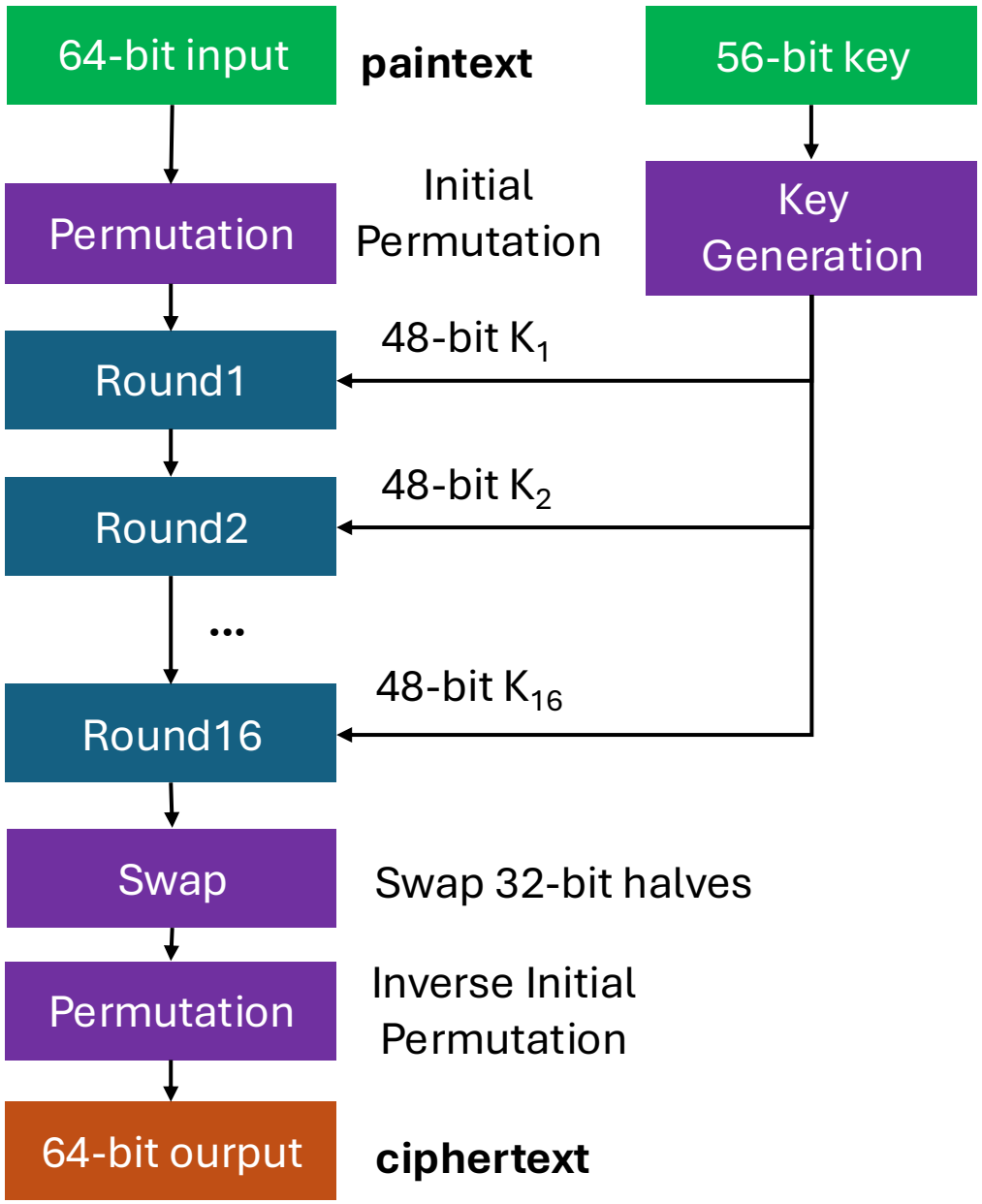



Image: https://en.wikipedia.org/wiki/Feistel_cipher

# DES Structure

- Input: **56-bit key** (plus 8 parity bits) and **64-bit plaintext**
- Output: **64-bit ciphertext**
- **16 rounds** of processing
- *Key generation:*
  - 56-bit key is permuted (PC-1), split into two 28-bit halves
  - For each round, bits are shifted and permuted (PC-2) to generate a unique **48-bit round key**



# DES Security Flaws

- A **56-bit key is too short** (Diffie-Hellman, 1977) → $7.2 * 10^{16}$ values
- In 1997, <u>3500 machines in parallel can solve a DES key in 4 months</u>
- In 1998, a special "DES" cracker was built for about $100,000 to <u>find a DES key in 4 days</u>
- DES has 4 **<u>weak keys</u>** since they generate the same sub-keys for every round

(1) 00000000 00000000 (2) 00000000 FFFFFFFF

(3) FFFFFFFF 00000000 (4) FFFFFFFF FFFFFFFF

# Triple DES (3-DES)

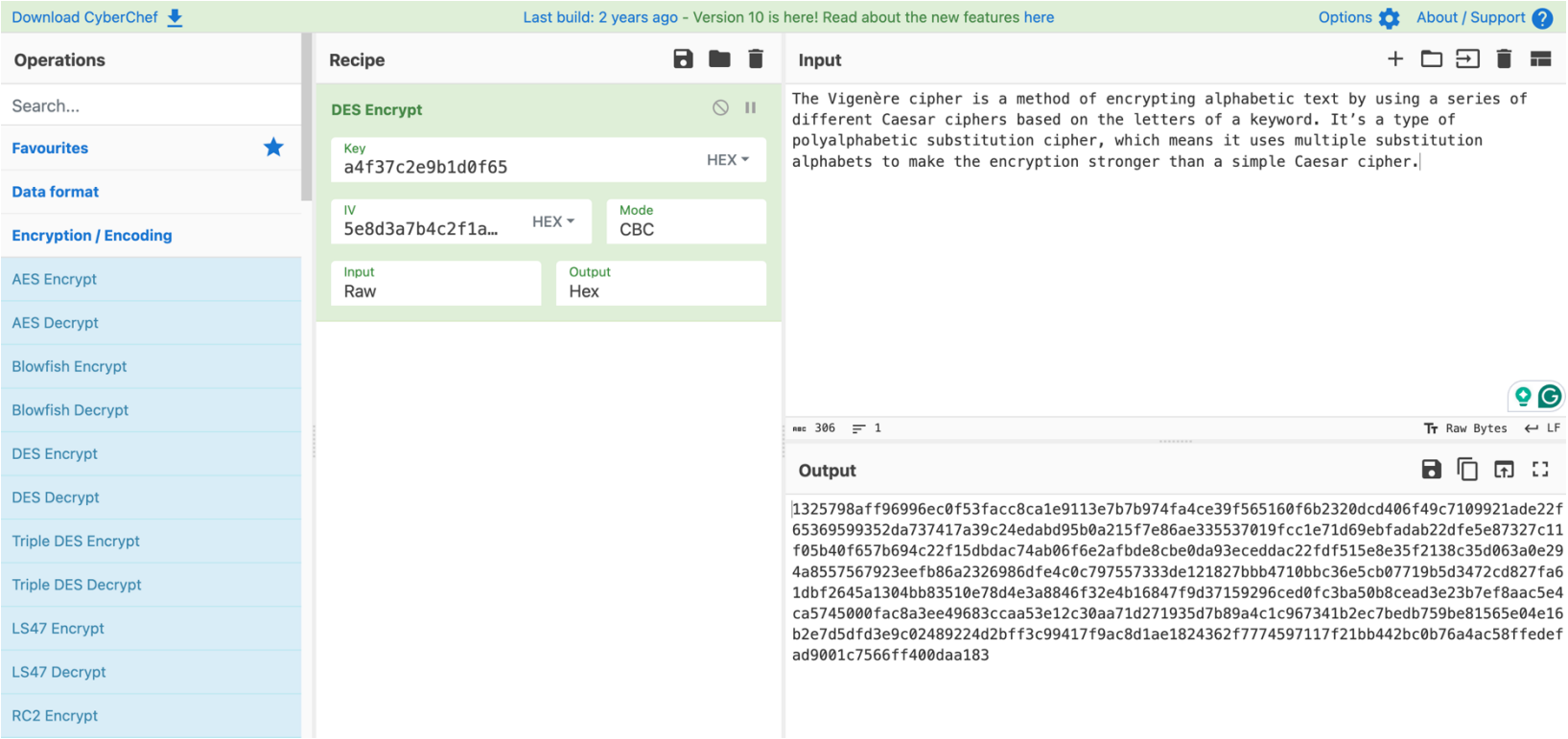
- Combine 3 DES using **3 keys**
- $C = E(K_3, D(K_2, E(K_1, P)))$
- Decryption is the same with keys reversed
- Triple DES with two keys: use K1 = K3

- **Strength:** increased security against brute-force attacks compared to DES
- **Weakness:** much slower than DES or AES, and small block size (64 bits)
- Superseded by AES

Image: https://www.rootstrap.com/blog/the-best-encryption-algorithms-for-cyber-security

# Try CyberChef!

- Web-based tool for performing encryption/decryption, cryptanlysis, encodind/decodeing
- https://gchq.github.io/CyberChef/



# Summary

- Basic terminologies: cryptography, cryptanalysis, plaintext, ciphertext
- Concepts of cryptography: encryption & decryption
- **Classical Ciphers:**
  - Substitution: Monoalphabetic, Polyalphabetic, OTP
  - Transposition: Rail Fence, Row Transposition
  - Product cipher
- **Modern Ciphers:**
  - Symmetirc-key encryption
  - DES & 3-DES

มหาวิทยาลัยมหิดล
Mahidol University

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